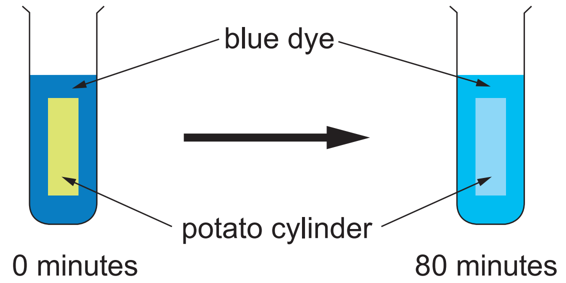
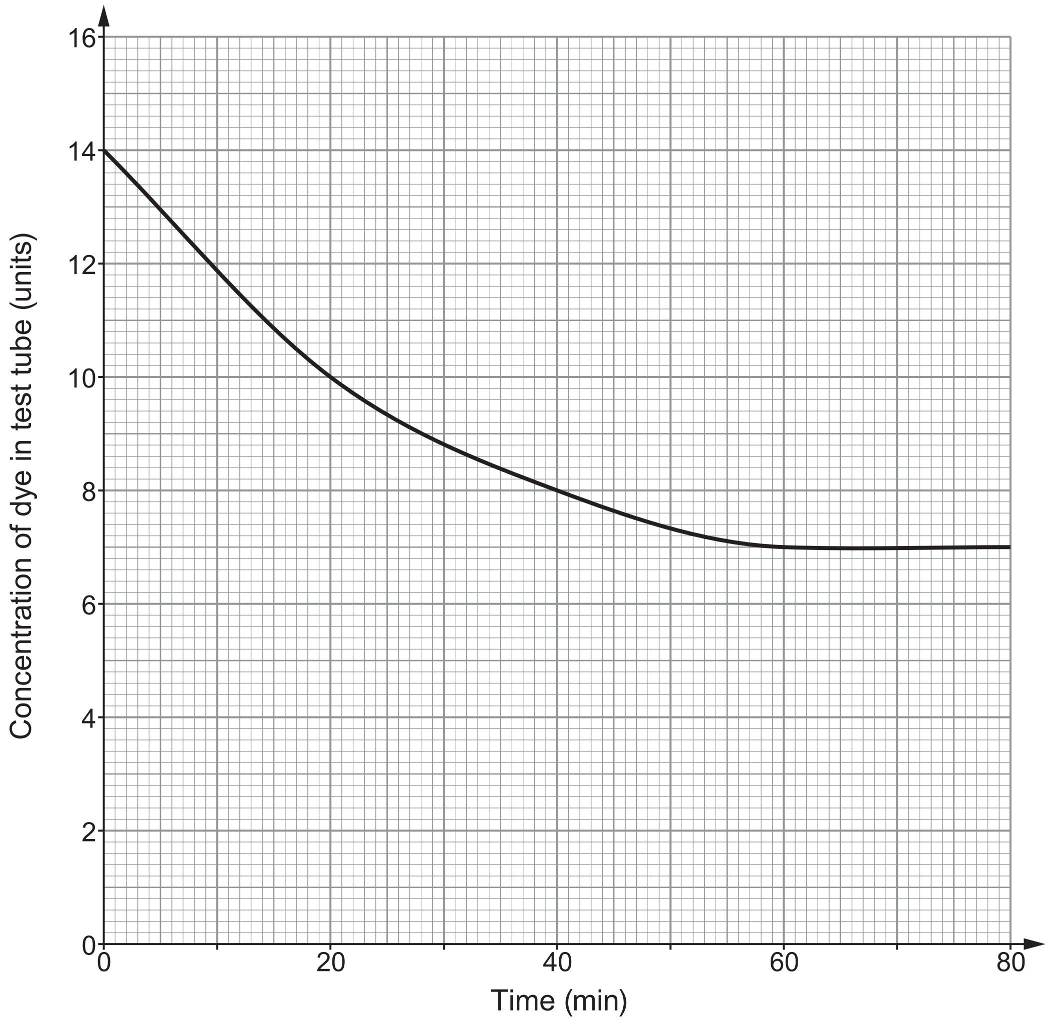
Examiner only

2. Several groups of students investigated the rate of diffusion of blue dye into potato by the method below.

1. Cut a cylinder of potato.
2. Place the cylinder in a test tube containing a solution of blue dye.
3. Put the test tube in a water bath at 20°C.
4. Measure the concentration of the dye after 20 minutes.
5. Continue to measure the concentration every 20 minutes up to 80 minutes.



The concentration of the dye left in the test tube after each 20-minute interval is shown on the graph below.



Examiner only

(a) (i) State the dependent variable in this experiment. [1]

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(ii) State **two** variables that should be controlled. [2]

1. ....

2. ....

(b) (i) State how the concentration of dye left in the test tube changed during the first 20 minutes. [1]

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(ii) Explain why this change occurred. [2]

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(c) One student suggests that the rate of diffusion is constant over the whole 80 minutes. Explain whether you agree. [3]

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(d) The students repeated the experiment in a water bath at 30°C. **Add another line** to the graph to show the expected results. [2]

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